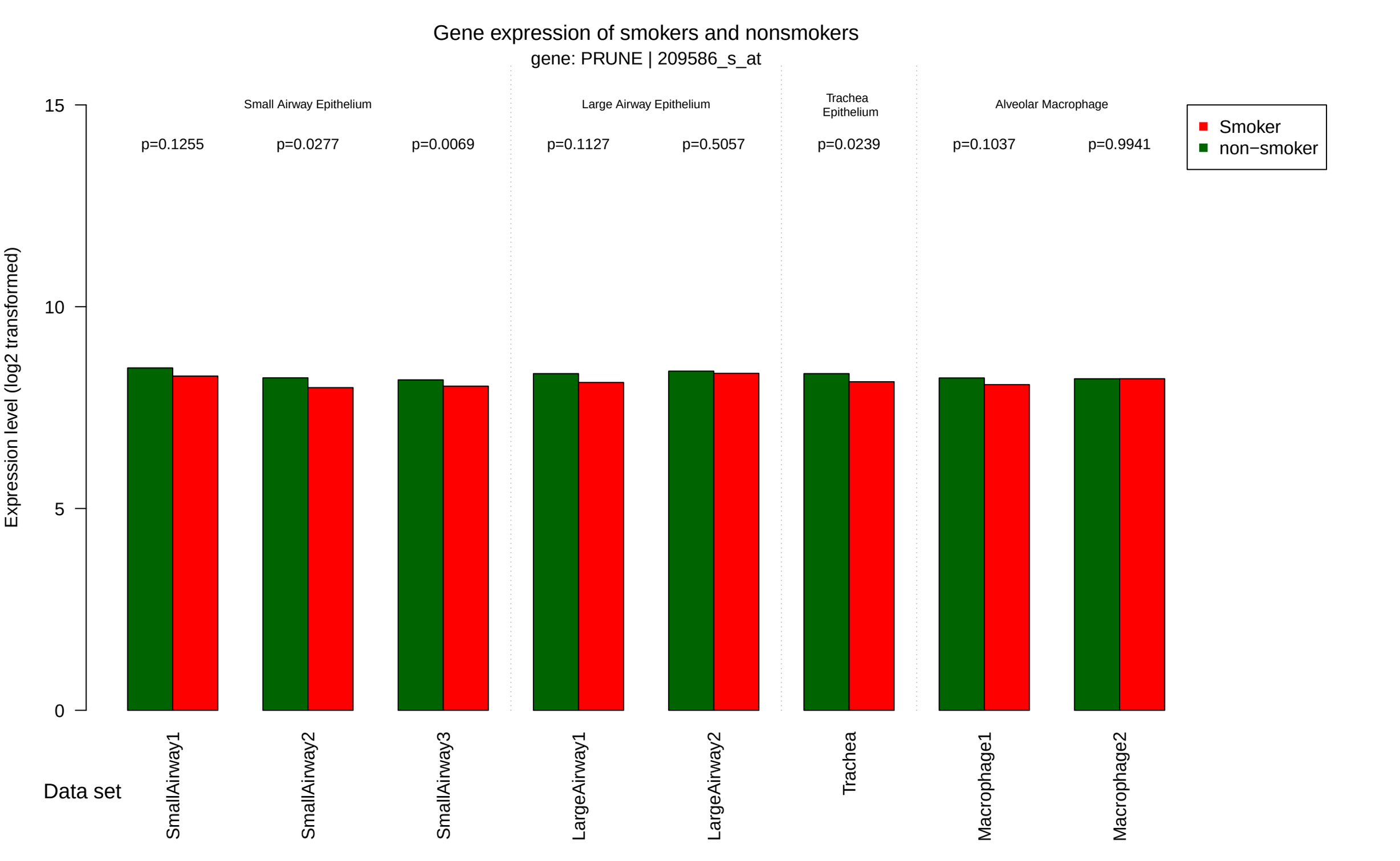
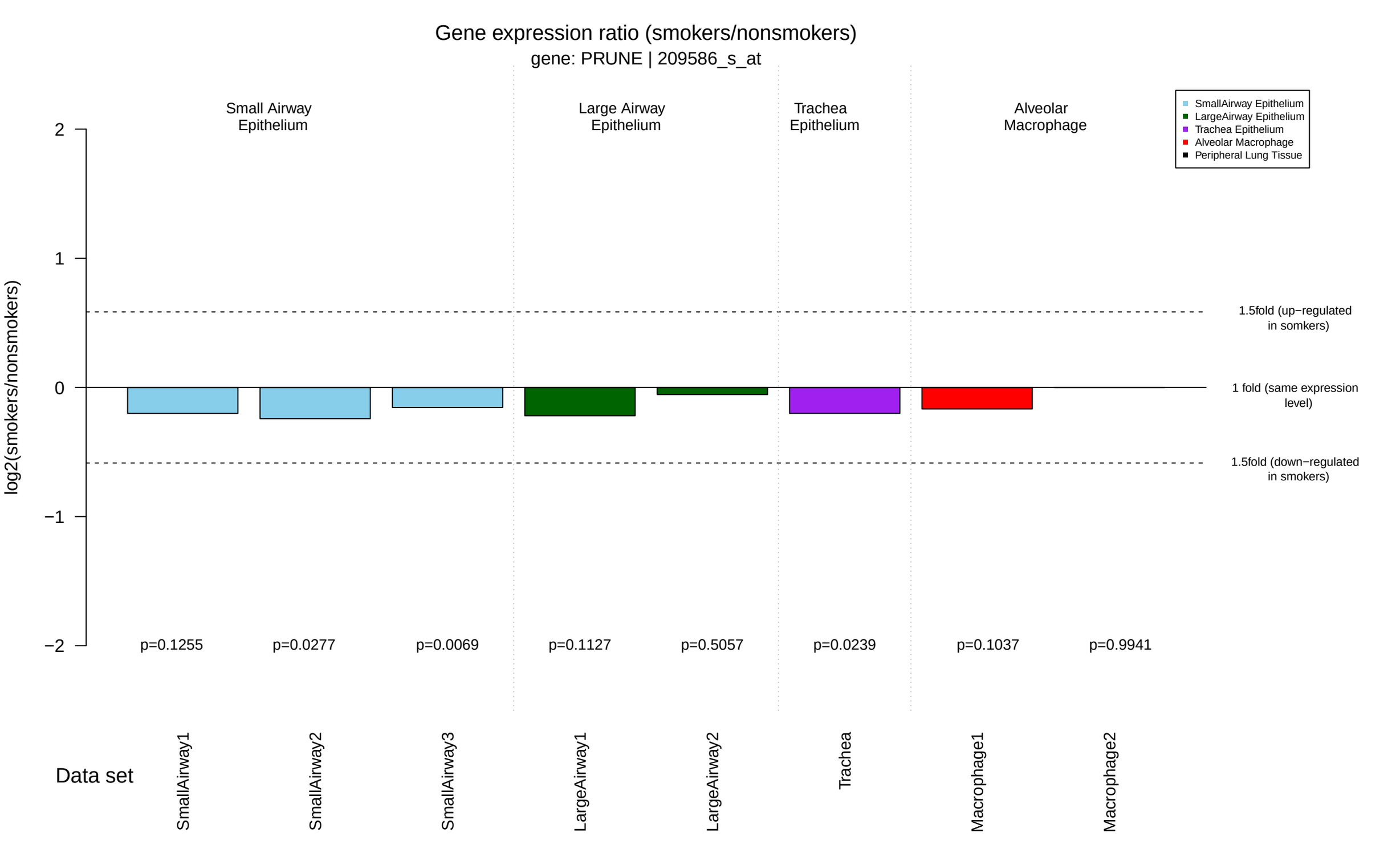
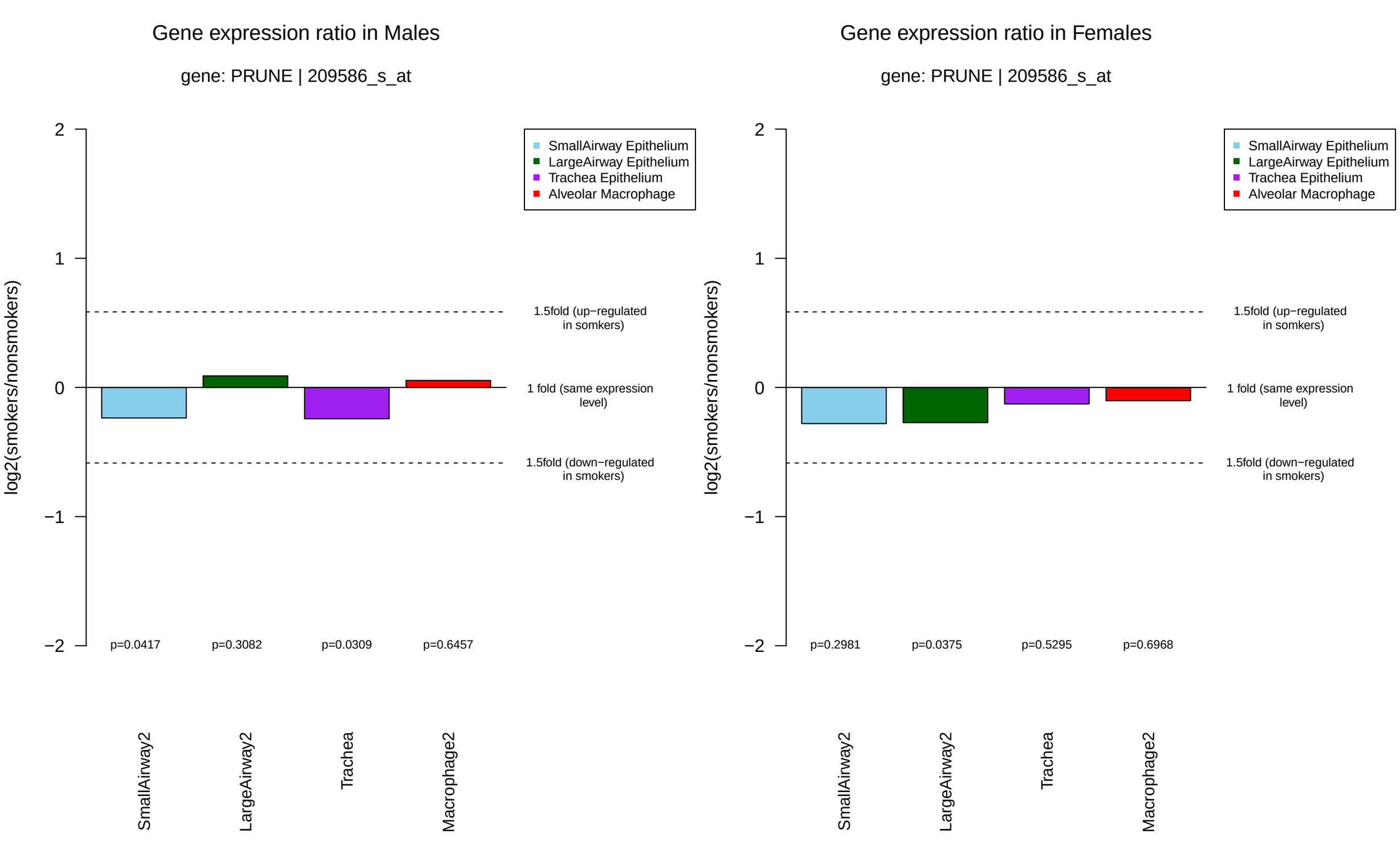
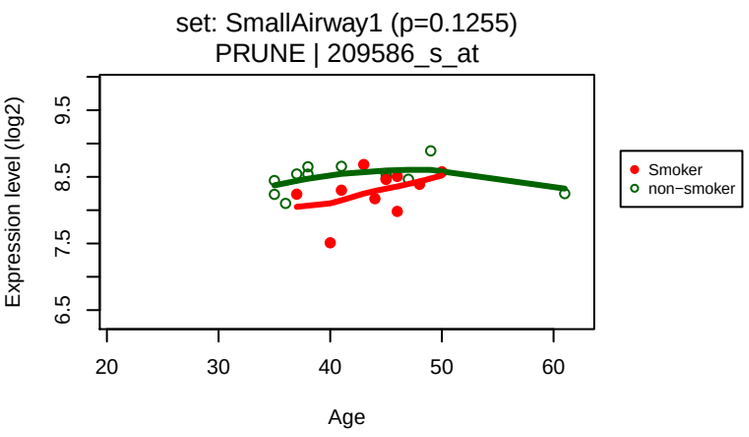
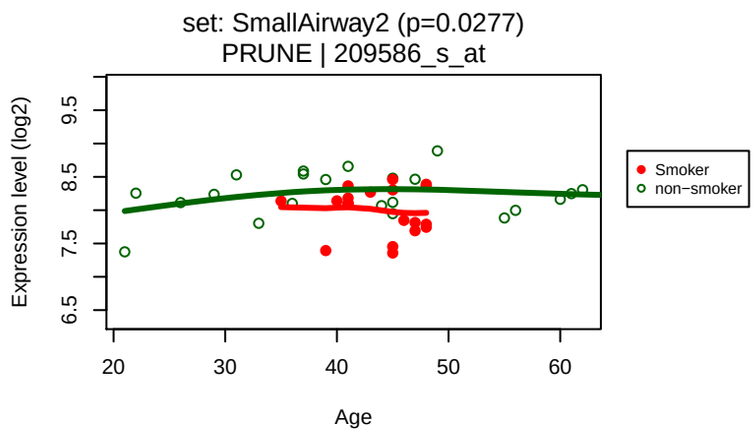
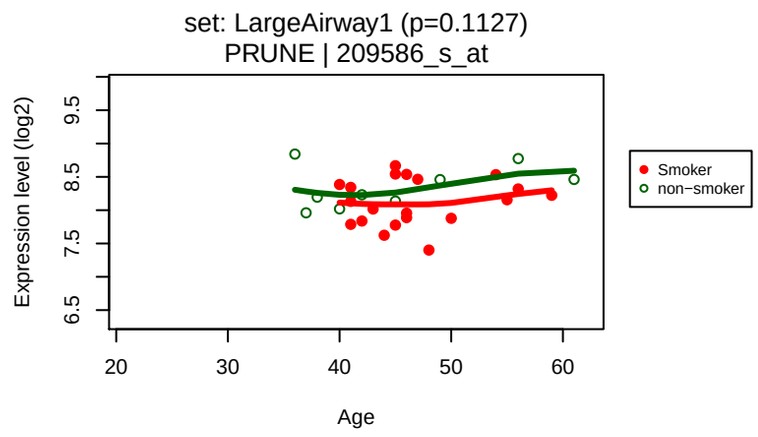

Gene expression of smokers and nonsmokers
gene: PRUNE | 209586_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1255
p=0.0277
p=0.0069
p=0.1127
p=0.5057
p=0.0239
p=0.1037
p=0.9941
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

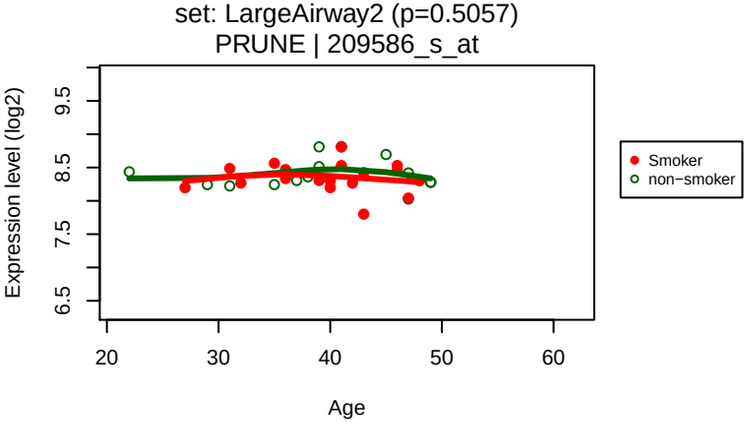

Gene expression ratio (smokers/nonsmokers)
gene: PRUNE | 209586_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1255
p=0.0277
p=0.0069
p=0.1127
p=0.5057
p=0.0239
p=0.1037
p=0.9941
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

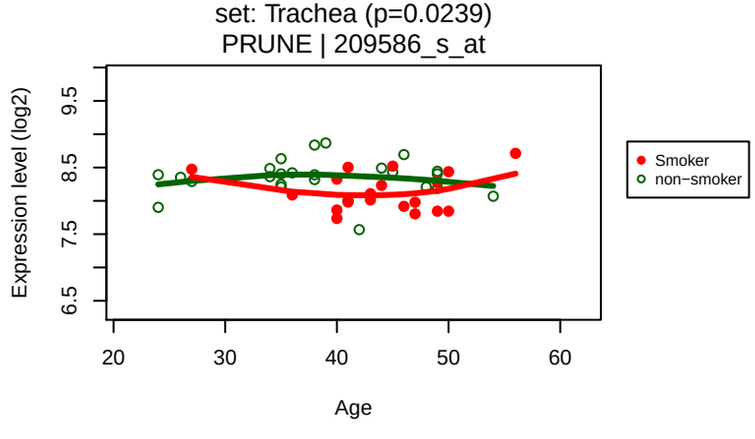

Gene expression ratio in Males
gene: PRUNE | 209586_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0417
p=0.3082
p=0.0309
p=0.6457
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRUNE | 209586_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2981
p=0.0375
p=0.5295
p=0.6968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

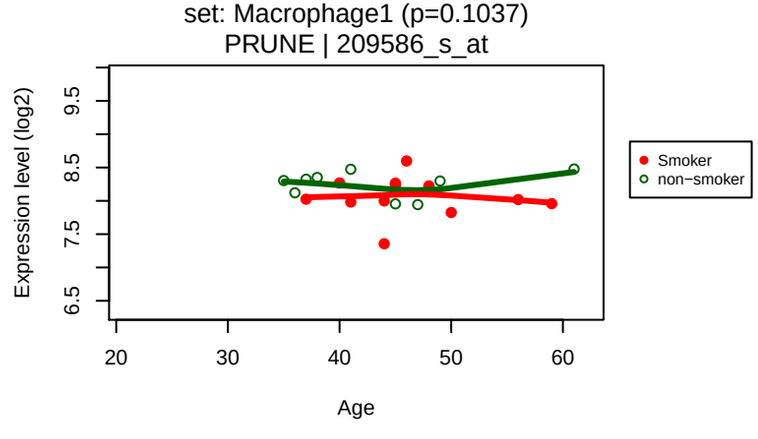

set: SmallAirway1 (p=0.1255)
PRUNE | 209586_s_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

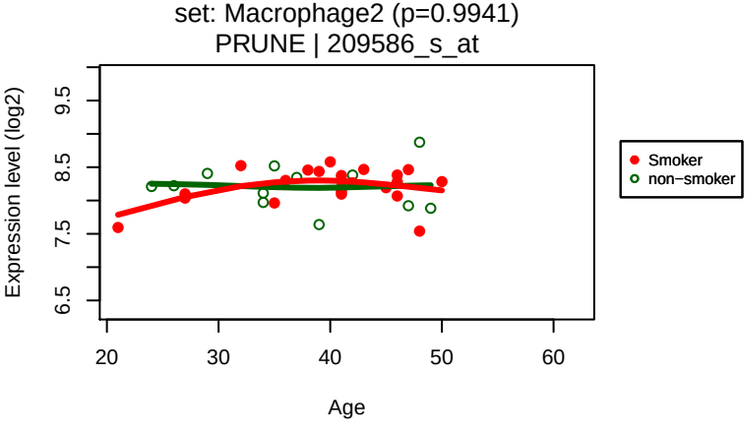

set: SmallAirway2 (p=0.0277)
PRUNE | 209586_s_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker

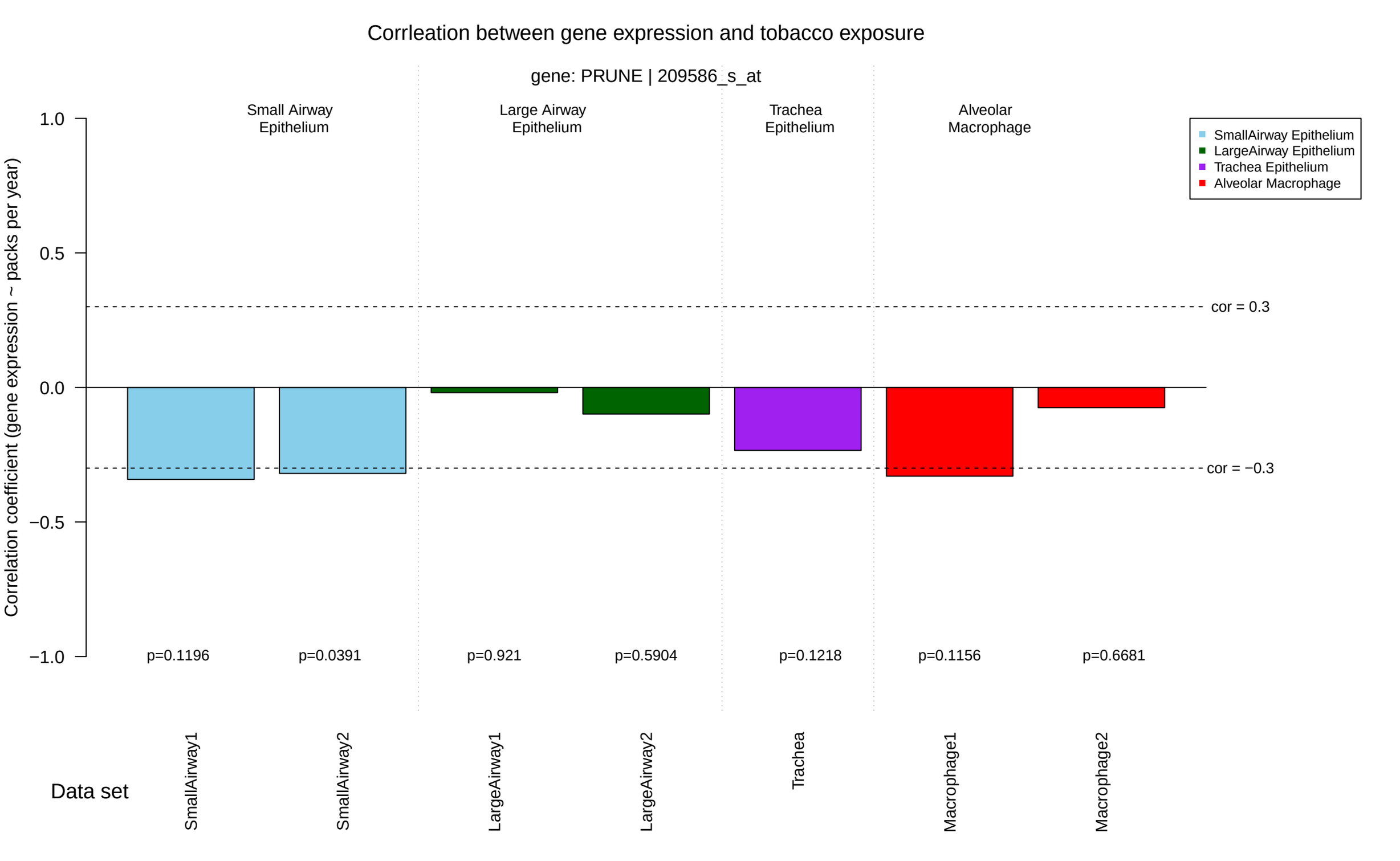

set: LargeAirway1 (p=0.1127)
PRUNE | 209586_s_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5057)
PRUNE | 209586_s_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0239)
PRUNE | 209586_s_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1037)
PRUNE | 209586_s_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9941)
PRUNE | 209586_s_at
Expression level (log2)
6.5
7.5
8.5
9.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRUNE | 209586_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1196
p=0.0391
p=0.921
p=0.5904
p=0.1218
p=0.1156
p=0.6681
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2